

Naifeng Zhang

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Research Focus: Mathematics-based code generation and reasoning for high-performance computational kernels.

EDUCATION

- 2021–27 **Carnegie Mellon University**
Ph.D. in Electrical and Computer Engineering
Advisor: Franz Franchetti
- 2021–24 **Carnegie Mellon University**
M.S. in Electrical and Computer Engineering
Advisor: Franz Franchetti
- 2017–21 **University of Southern California**
B.S. in Computer Science
Advisor: Viktor K. Prasanna
Thesis: Lightweight Augmented Neural Network for Performance Prediction and Its Applications
W.V.T. Rusch Undergraduate Engineering Honors Program
- 2017–21 **University of Southern California**
B.S. in Mathematics
Departmental Honors Program

EXPERIENCE

- 2021– **Carnegie Mellon University, SPIRAL Group**
Graduate Researcher, Pittsburgh, PA
- Developing code generation systems targeting CPUs [C4, W3], GPUs [M4, C6, W8], and specialized hardware [C7, C9] for high-performance scientific and cryptographic computing.
 - Developing semantics lifting techniques for program reasoning, enabling code modernization and verification of AI-generated scientific code [W1, W2].
- 2026 **NVIDIA, CUDA Math Libraries**
Intern, Remote
- Prototyping a code-generation framework for CUDA Tile implementations of high-performance math library kernels.
 - Selected for the cuFFT team’s quarterly “Top-5 Things.”
- 2025 **NVIDIA, Programming Systems and Applications Research Group**
Intern, Santa Clara, CA
- Extended [C6] with architecture-aware optimizations for NVIDIA cryptographic workloads.
 - Achieved near-peak performance across multiple NVIDIA GPUs [M4]; selected for NVIDIA Research’s quarterly “Top-5 Things.”
- 2018–21 **University of Southern California, Data Science Lab**
Undergraduate Researcher, Los Angeles, CA
- Developed lightweight neural-network-based performance prediction methods [C11] and a prediction-guided auto-tuner [C10] for heterogeneous platforms.

GRANTS

- 2023- **NSF ACCESS**, *High-Performance Code Generation for Homomorphic Encryption on GPUs using SPIRAL*
N. Zhang (PI), F. Franchetti (Co-PI)
- Awarded 200,000 ACCESS Credits for tuning, benchmarking, and extending SPIRAL-generated cryptographic kernels for homomorphic encryption on state-of-the-art CPUs, GPUs, and wafer-scale engines.

AWARDS

- 2026 **Third Place, ACM Student Research Competition**, PLDI
For “Semantics Lifting for Scientific Kernels”
- 2024 **Best Poster Runner-up**, PRISM Annual Review, Systems & Software Track
For “Accelerating High-Precision Number Theoretic Transforms using Intel AVX-512”
- 2023 **Outstanding Short Paper Award**, HPEC
For “Twiddle Factor Generation for a Vectorized Number Theoretic Transform”
- 2023 **Second Place, ACM Student Research Competition**, CGO
For “Generating Number Theoretic Transforms for Multi-Word Integer Data Types”
- 2021 **Carnegie Institute of Technology Dean’s Fellow**, Carnegie Mellon University
- 2021 **Discovery Scholar Distinction**, University of Southern California
- 2019–21 **Provost’s Research Fellowship**, University of Southern California
- 2018–21 **Academic Achievement Award**, University of Southern California

PUBLICATIONS

Underlined names denote undergraduate and Master’s mentees.

Conference Proceedings

1. S. Rao, A. Prakash, **N. Zhang**, H. Mankad, F. Franchetti. “LibraryX: A Framework for Cross-Library-Call Optimization.” The IEEE International Parallel & Distributed Processing Symposium (**IPDPS**), 2026.
2. **N. Zhang**, S. McAleer, T. Sandholm. “Faster Game Solving via Hyperparameter Schedules.” The AAAI Conference on Artificial Intelligence (**AAAI**), 2026.
3. Y. Lan, L. Tang, **N. Zhang**, Y. Eum, J. Hoe, F. Franchetti. “A RISC-V Vector Extension for Multi-word Arithmetic.” The International Workshop on RISC-V for HPC (**RISCV-HPC**), in conjunction with the International Conference for High Performance Computing, Networking, Storage, and Analysis (**SC**), 2025.
4. **N. Zhang**, S. Fu, F. Franchetti. “Towards Closing the Performance Gap for Cryptographic Kernels Between CPUs and Specialized Hardware.” The IEEE/ACM International Symposium on Microarchitecture (**MICRO**), 2025.
5. Q. Oschatz, **N. Zhang**, M. Franusich, F. Franchetti. “Towards Automated Reasoning Chains for Verification of LLM-Generated Scientific Code.” IEEE High Performance Extreme Computing Conference (**HPEC**), 2025.
6. **N. Zhang**, F. Franchetti. “Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic on GPU.” The International Symposium on Code Generation and Optimization (**CGO**), 2025.

7. **N. Zhang**, A. Ebel, N. Neda, P. Brinich, B. Reynwar, A. G. Schmidt, M. Franusich, J. Johnson, B. Reagen, F. Franchetti. “Generating High-Performance Number Theoretic Transform Implementations for Vector Architectures.” IEEE High Performance Extreme Computing Conference (**HPEC**), 2023.
8. D. Sun, **N. Zhang**, F. Franchetti. “Optimization and Performance Analysis of Shor’s Algorithm in Qiskit.” IEEE High Performance Extreme Computing Conference (**HPEC**), 2023.
9. D. Soni, N. Neda, **N. Zhang**, B. Reynwar, H. Gamil, B. Heyman, M. N. T. Moopan, A. Al Badawi, Y. Polyakov, K. Canida, M. Pedram, M. Maniatakos, D. B. Cousins, F. Franchetti, M. French, A. Schmidt, B. Reagen. “RPU: The Ring Processing Unit.” IEEE International Symposium on Performance Analysis of Systems and Software (**ISPASS**), 2023.
10. **N. Zhang**, A. Srivastava, R. Kannan, V. K. Prasanna. “GenMAT: A General-Purpose Machine Learning-Driven Auto-Tuner for Heterogeneous Platforms.” The Workshop on Programming Environments for Heterogeneous Computing (**PEHC**), in conjunction with the International Conference for High Performance Computing, Networking, Storage, and Analysis (**SC**), 2021.
11. A. Srivastava*, **N. Zhang***, R. Kannan, V. K. Prasanna. “Towards High Performance, Portability, and Productivity: Lightweight Augmented Neural Networks for Performance Prediction.” The International Conference on High Performance Computing, Data, and Analytics (**HiPC**), 2020. **Equal contribution.*
12. C. Imes, A. Colin, **N. Zhang**, A. Srivastava, V. K. Prasanna, J. P. Walters. “Compiler Abstractions and Runtime for Extreme-scale SAR and CFD Workloads.” The Workshop on Extreme Scale Programming Models and Middleware (**ESPM2**), in conjunction with the International Conference for High Performance Computing, Networking, Storage, and Analysis (**SC**), 2020.

Refereed Workshop Papers and Extended Abstracts

1. **N. Zhang**, F. Franchetti. “Semantics Lifting for Scientific Kernels.” The ACM SIGPLAN Conference on Programming Language Design and Implementation (**PLDI**), 2026. **Third Place, ACM Student Research Competition.**
2. **N. Zhang**, S. Rao, M. Franusich, F. Franchetti. “Towards Semantics Lifting for Scientific Computing: A Case Study on FFT.” The Theory and Practice of Static Analysis Workshop (**TPSA**), in conjunction with the ACM SIGPLAN Symposium on Principles of Programming Languages (**POPL**), 2025.
3. S. Fu, **N. Zhang**, F. Franchetti. “Accelerating High-Precision Number Theoretic Transforms using Intel AVX-512.” The International Conference on Parallel Architectures and Compilation Techniques (**PACT**), 2024. **First Place, ACM Student Research Competition. Best Poster Runner-up** at PRISM Annual Review, Systems & Software Track.
4. Y. Eum, **N. Zhang**, L. Tang, F. Franchetti. “Towards a RISC-V Instruction Set Extension for Multi-word Arithmetic.” IEEE High Performance Extreme Computing Conference (**HPEC**), 2024.
5. P. Brinich, **N. Zhang**, A. Ebel, F. Franchetti, J. Johnson. “Twiddle Factor Generation for a Vectorized Number Theoretic Transform.” IEEE High Performance Extreme Computing Conference (**HPEC**), 2023. **Outstanding Short Paper Award.**
6. H. Mankad, A. Rovinelli, M. Zecevic, P. McCorquodale, F. Franchetti, **N. Zhang**, S. Rao, R. A. Lebensohn, L. Capolungo. “EVPFFTX: A First Look at FFTX Applications in Material Science.” IEEE High Performance Extreme Computing Conference (**HPEC**), 2023.
7. D. B. Cousins, Y. Polyakov, A. Al Badawi, M. French, A. Schmidt, A. Jacob, B. Reynwar, K. Canida, A. Jaiswal, C. Mathew, H. Gamil, N. Neda, D. Soni, M. Maniatakos, B. Reagen, **N. Zhang**, F. Franchetti, P. Brinich, J. Johnson, P. Broderick, M. Franusich, B. Zhang, Z. Cheng, M. Pedram. “TREBUCHET: Fully Homomorphic Encryption Accelerator for Deep Computation.” The Government Microcircuit Applications and Critical Technology Conference (**GOMACTech**), 2023.

8. **N. Zhang**, F. Franchetti. “Generating Number Theoretic Transforms for Multi-Word Integer Data Types.” The International Symposium on Code Generation and Optimization (**CGO**), 2023. **Second Place, ACM Student Research Competition**.
9. **N. Zhang**, H. Gamil, P. Brinich, B. Reynwar, A. Al Badawi, N. Neda, D. Soni, K. Canida, Y. Polyakov, P. Broderick, M. Maniatakos, A. G. Schmidt, M. Franusich, J. Johnson, B. Reagen, D. B. Cousins, F. Franchetti. “Towards Full-Stack Acceleration for Fully Homomorphic Encryption.” IEEE High Performance Extreme Computing Conference (**HPEC**), 2022.
10. I. Grosz, **N. Zhang**, M. Heule. “Towards the shortest DRAT proof of the Pigeonhole Principle.” The Pragmatics of SAT Workshop (**PoS**), in conjunction with the International Conference on Theory and Applications of Satisfiability Testing (**SAT**), 2022.

Manuscripts Under Review

1. S. Alsultan, P. Brinich, **N. Zhang**, J. Johnson. “Performance Analysis of Vectorized Number Theoretic Transforms.”
2. Y. Lan, L. Tang, **N. Zhang**, F. Franchetti. “Exploring FP64-Equivalent FFT Execution on AWS Trainium via Software-Emulated Floating-Point Arithmetic.”
3. Y. Lan, L. Tang, **N. Zhang**, F. Franchetti. “Efficient High-Precision Floating-Point Arithmetic Using Integer Units: A First Look.”
4. **N. Zhang**, C. Gouert, S. Dalton, V. Joseph, M. Garland, F. Franchetti. “Architecture-Aware Code Generation for Number Theoretic Transforms on GPUs.”

TALKS

Invited Talks

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| 2026 | <i>Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic</i>
The Cryptolets Tutorial, in conjunction with the International Symposium on Computer Architecture (ISCA), Raleigh, NC, Jun. 28 |
| 2026 | <i>Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic</i>
Carnegie Mellon University , Graduate Student and Postdoc Seminars (GSPS), Department of Mathematical Sciences, Feb. 26 |
| 2026 | <i>Closing the Performance Gap for Cryptographic Kernels via Code Generation</i>
University of California, Los Angeles , VLSI Architecture, Synthesis, and Technology (VAST) Laboratory, Jan. 27 |
| 2025 | <i>Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic</i>
The Workshop on Architectures for Zero-Knowledge Proofs and Verifiable Computation (ZKARCH), in conjunction with the IEEE/ACM International Symposium on Microarchitecture (MICRO), Virtual, Oct. 18 |
| 2025 | <i>Towards Closing the Performance Gap for Cryptographic Kernels Between CPUs and Specialized Hardware</i>
Carnegie Mellon University , Computer Architecture Lab at Carnegie Mellon (CALCM), Oct. 2 |
| 2025 | <i>Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic</i>
University of Southern California , Ming Hsieh Department of Electrical and Computer Engineering, May 9 |

- 2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
New York University, Department of Electrical and Computer Engineering, May 2
- 2025 *Optimization and Performance Analysis of Shor's Algorithm in Qiskit and Beyond*
Carnegie Mellon University, The Center for Quantum Computing and Information Technologies (QCiT), Apr. 1
- 2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
University of Pennsylvania, The Programming Languages Group at the University of Pennsylvania (PLClub), Feb. 21
- 2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
Carnegie Mellon University, Computer Architecture Lab at Carnegie Mellon (CALCM), Feb. 14

Guest Lectures

- 2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
Carnegie Mellon University, Computational Problem Solving for Engineers, Apr. 1

Conference Presentations

- 2026 *Semantics Lifting for Performance Portability*
The Society for Industrial and Applied Mathematics (**SIAM**) Annual Meeting, Cleveland, OH, Jul. 7
- 2026 *Semantics Lifting for Scientific Kernels*
The ACM SIGPLAN Conference on Programming Language Design and Implementation (**PLDI**), Boulder, CO, Jun. 18
- 2026 *Semantics Lifting for Quantum Circuits*
Penn Forum on Quantum Systems (**Penn FoQuS**), Philadelphia, PA, Apr. 21
- 2025 *Towards Closing the Performance Gap for Cryptographic Kernels Between CPUs and Specialized Hardware*
The IEEE/ACM International Symposium on Microarchitecture (**MICRO**), Virtual, Oct. 22
- 2025 *Towards Semantics Lifting for Scientific Computing: A Case Study on FFT*
The **ORNL AI4Science** Workshop, Oak Ridge, TN, Apr. 30
- 2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic on GPU*
The International Symposium on Code Generation and Optimization (**CGO**), Las Vegas, NV, Mar. 4
- 2025 *Towards Semantics Lifting for Scientific Computing: A Case Study on FFT*
The Theory and Practice of Static Analysis Workshop (**TPSA**), in conjunction with the ACM SIGPLAN Symposium on Principles of Programming Languages (**POPL**), Denver, CO, Jan. 21
- 2023 *Generating High-Performance Number Theoretic Transform Implementations for Vector Architectures*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 29
- 2023 *Generating Number Theoretic Transforms for Multi-Word Integer Data Types*
The International Symposium on Code Generation and Optimization (**CGO**), Montreal, Canada, Feb. 28
- 2022 *Towards Full-Stack Acceleration for Fully Homomorphic Encryption*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 23
- 2021 *GenMAT: A General-Purpose Machine Learning-Driven Auto-Tuner for Heterogeneous Platforms*
The Workshop on Programming Environments for Heterogeneous Computing (**PEHC**), in conjunction with the International Conference for High Performance Computing, Networking, Storage, and Analysis (**SC**), Virtual, Nov. 19

- 2020 *Towards High Performance, Portability, and Productivity: Lightweight Augmented Neural Networks for Performance Prediction*
The International Conference on High Performance Computing, Data, and Analytics (**HiPC**), Virtual, Dec. 16

TEACHING

Carnegie Mellon University

Teaching assistant

- 2024 Mathematical Foundations of Electrical Engineering
2023 Computational Problem Solving for Engineers

University of Southern California

Undergraduate teaching assistant

- 2021 Special Topics: Accelerated Computing Using FPGAs
2020 Parallel and Distributed Computation
2020 Special Topics: Accelerated Computing Using FPGAs
2020 Discrete Methods in Computer Science
2019 Parallel and Distributed Computation
2019 Discrete Methods in Computer Science

Conference Tutorials

Presenter

- 2026 *SPIRAL Tutorial: A Code Generation Approach to Hardware-Software Co-Design*
ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), Pittsburgh, PA, Mar. 23
2025 *Open Source SPIRAL 8.5.1 Tutorial*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 17
2024 *Open Source SPIRAL 8.5 Tutorial*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 25
2023 *Open Source SPIRAL 8.5 Tutorial*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 27

MENTORING

All mentees are from Carnegie Mellon University unless otherwise noted. Parentheses denote first known post-graduation affiliation.

Undergraduate

2025 Yiwen Jiang
2024–26 Sophia Fu (Apple)
2024–25 Misho Alexandrov
2024 Zubin Narayan
2024 Youngjin Eum
2024 Govind Malasani
2024 Steven Lee (Stripe)
2023–25 Gordon Xu (Etched)
2022–23 Matt Ngaw (Arm)
2022–23 Jimmy Zhou (NVIDIA)

Master's

2025–26 Yunhao Lan (CMU Ph.D.)
2024–25 Yujun Lee (Meta)
2023 Kofi Poku (PNNL)
2022–23 Dewang Sun (Cadence)
2022 Hongbo Sun (Alibaba)

SERVICE

Conference Program Organization

Co-organizer, Minisymposium on *Performance Portability and Scalability for Scientific Applications* at the Society for Industrial and Applied Mathematics (**SIAM**) Annual Meeting, 2026

Co-organizer, Tutorial on *SPIRAL: A Code Generation Approach to Hardware-Software Co-Design* at the ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), 2026

Program Committees

The AAAI Conference on Artificial Intelligence (**AAAI**), 2026–27

The Workshop on Systems and Architectures for Encrypted AI (**SAFE-AI**), in conjunction with the International Symposium on Computer Architecture (**ISCA**), 2026

The Workshop on AI Assisted Software Development for HPC (**AI4Dev**), in conjunction with the International Conference on Parallel Processing (**ICPP**), 2025

Journal Peer Review

IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**), 2025–26

IEEE Transactions on Dependable and Secure Computing (**TDSC**), 2025–26

IEEE Transactions on Parallel and Distributed Systems (**TPDS**), 2025–26

IEEE Transactions on Mobile Computing (**TMC**), 2025–26

IEEE Transactions on Computational Social Systems (**TCSS**), 2025

IEEE Transactions on Emerging Topics in Computing (**TETC**), 2025

ACM Computing Surveys (**CSUR**), 2025

The International Journal of High Performance Computing Applications (**IJHPCA**), 2025

IEEE Transactions on Information Forensics & Security (**T-IFS**), 2025

Service to the University

Graduate Student Representative, College of Engineering Council, Carnegie Mellon University, 2025–26

Member, Department of Electrical and Computer Engineering Faculty Hiring Student Council, Carnegie Mellon University, 2022–25

Outreach

Member, College of Engineering Graduate Student Outreach Committee, Carnegie Mellon University, 2023

MEDIA

Naifeng Zhang: Researching High-Speed Computational Kernels, College of Engineering, Carnegie Mellon University, 2026

Updated August 2026